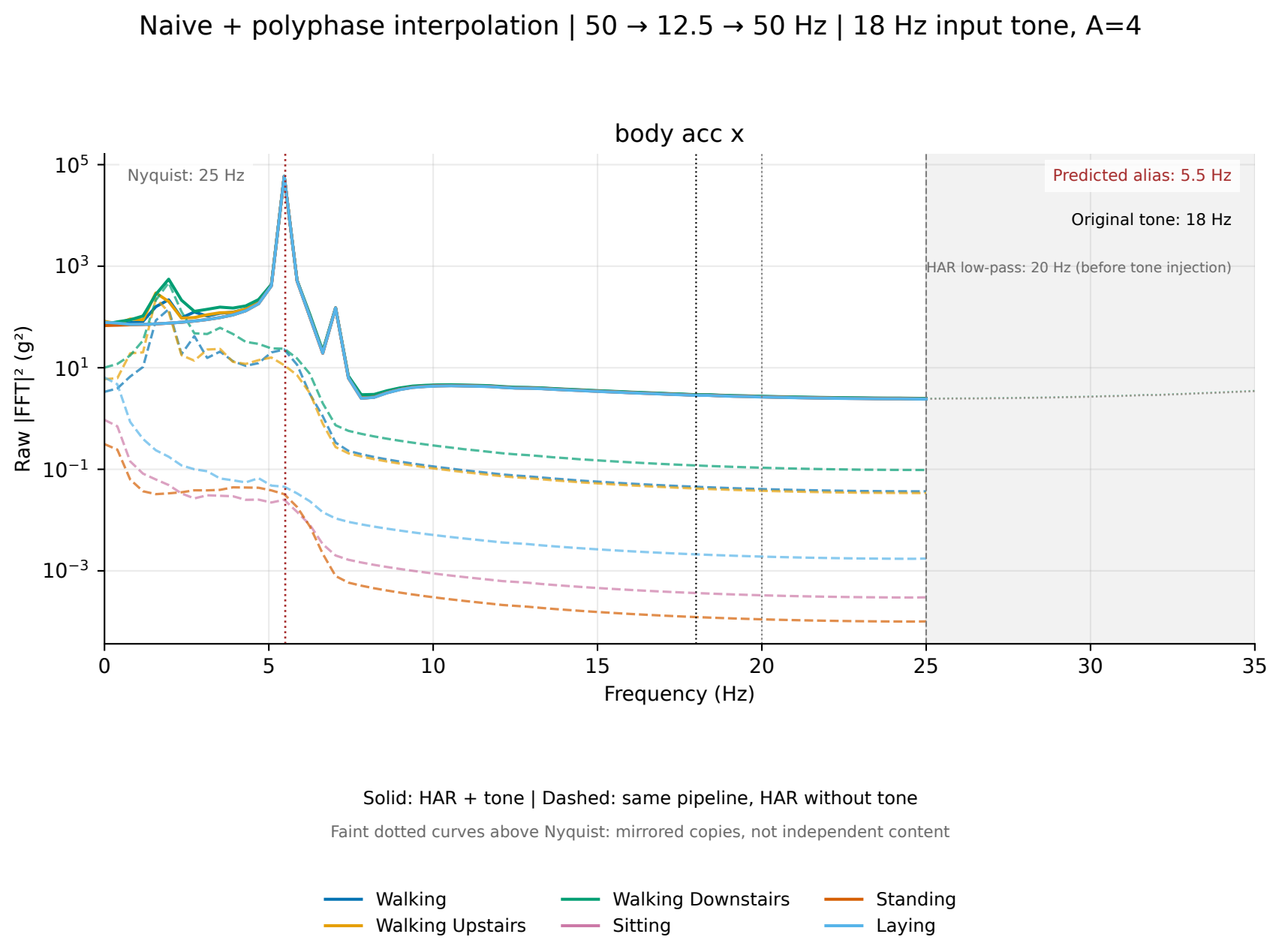
Naive + polyphase interpolation | 50 → 12.5 → 50 Hz | 18 Hz input tone, A=4
body acc x
Nyquist: 25 Hz
Predicted alias: 5.5 Hz
Original tone: 18 Hz
HAR low-pass: 20 Hz (before tone injection)
Raw |FFT|² (g²)
Frequency (Hz)
Solid: HAR + tone | Dashed: same pipeline, HAR without tone
Faint dotted curves above Nyquist: mirrored copies, not independent content
Walking
Walking Upstairs
Walking Downstairs
Sitting
Standing
Laying